

Computational modeling of UHF/SHF electromagnetic field propagation and intensity in enclosures with dielectric materials

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Abstract

The study of the spatial distribution of indoor electromagnetic waves is a fundamental problem in applied electromagnetism due to attenuation, dielectric absorption, and shielding phenomena caused by building structures. In this work, a two-dimensional computational model is developed to simulate stationary propagation and electromagnetic field intensity in the UHF ($2.4\ \mathrm{GHz}$, $\lambda = 12.5\ \mathrm{cm}$) and SHF ($5.0\ \mathrm{GHz}$, $\lambda = 6.0\ \mathrm{cm}$) bands within enclosures containing dielectric and conductive materials. The spatial domain is discretized using the centered finite-difference method, applying Dirichlet boundary conditions at radiating sources and absorbing boundaries, while incorporating local loss coefficients based on the relative permittivity (ε_r) and conductivity (σ) of each material (drywall, brick, concrete, and metal). The resulting algebraic system is iteratively solved via Successive Over-Relaxation (SOR), determining at each node the received power (dBm), the time-averaged Poynting vector magnitude ($\mu\mathrm{W/m^2}$), and the root-mean-square electric field E_{rms} ($\mathrm{V/m}$). Validation against the analytical log-distance and ITU-R P.1238 models yields a high correlation ($R^2 = 0.978$) and a root-mean-square error of $1.84\ \mathrm{dB}$, confirming the higher dielectric penetration of the $2.4\ \mathrm{GHz}$ band and the Faraday cage shielding effect in metallic enclosures.

Keywords: *Electromagnetic Waves; Dielectric Attenuation; Poynting Vector; Finite Differences; Successive Over-Relaxation (SOR).* | **Modality:** *Computational.*

Introduction and motivation

When a microwave electromagnetic wave propagates inside an enclosed space, its energy flux density decreases due to both geometric wavefront divergence and dissipation within walls characterized by relative permittivity ε_r and effective conductivity $\sigma > 0$ [1, 2]. While closed-form free-space analytical equations are only valid in homogeneous media, numerical modeling enables solving realistic architectural geometries.

General objective: To numerically model and validate the spatial distribution of received power (P_{rx}), time-averaged Poynting vector magnitude ($\langle S \rangle$), and root-mean-square electric field intensity (E_{rms}) in the $2.4\ \mathrm{GHz}$ and $5.0\ \mathrm{GHz}$ bands inside 2D enclosures with dielectric and conductive materials using finite differences and SOR relaxation.

Theoretical framework / Physical model

In air (intrinsic impedance $\eta_0 = \sqrt{\mu_0/\varepsilon_0} \approx 376.7\,\Omega$), the time-averaged magnitude of the **Poynting vector** $\langle S \rangle$ ($\mathrm{W/m^2}$) relates to the RMS electric field E_{rms} ($\mathrm{V/m}$), the RMS magnetic field H_{rms} ($\mathrm{A/m}$), and the received power P_{rx} captured by an isotropic effective aperture $A_{\mathrm{eff}} = \lambda^2/(4\pi)$ as follows [1]:

$$\langle S \rangle = \frac{E_{\mathrm{rms}}^2}{\eta_0} = \frac{4\pi\, P_{\mathrm{rx}}}{\lambda^2} \implies E_{\mathrm{rms}} = \sqrt{\frac{4\pi\,\eta_0\, P_{\mathrm{rx}}}{\lambda^2}}, \quad H_{\mathrm{rms}} = \frac{E_{\mathrm{rms}}}{\eta_0} \tag{1}$$

When traversing a lossy dielectric material or a conductor, the field attenuates exponentially according to the Beer-Lambert law $E(x) = E_0 e^{-\alpha x}$. In a 2D stationary regime, the normalized energy intensity envelope $u(x, y) \in [0, 100]$ over the domain Ω with spatial absorption $\kappa^2(x, y)$ is governed by the elliptic partial differential equation [3]:

$$\nabla^2 u(x, y) - \kappa^2(x, y)\, u(x, y) = 0 \quad \text{in } \Omega \tag{2}$$

subject to fixed **Dirichlet** boundary conditions at the radiating source $u(x_0, y_0) = u_{\mathrm{EIRP}}$ and null absorbing boundaries along the outer perimeter $u|_{\partial\Omega} = 0$.

Description of the modeled system

The developed software (*SimuNet Física*) discretizes $12\ \mathrm{m} \times 12\ \mathrm{m}$ enclosures into 24×24 node grids ($h = 0.5\ \mathrm{m}$), allowing placement of sources operating at $2.4\ \mathrm{GHz}$ ($\lambda = 12.5\ \mathrm{cm}$) and $5.0\ \mathrm{GHz}$ ($\lambda = 6.0\ \mathrm{cm}$) as well as walls with the constitutive properties listed in Table 1 [4].

Table 1. Electromagnetic properties and attenuation losses of modeled media (ITU-R P.2040).

Material / Medium	Permittivity ε_r	Conductivity σ (S/m)	Loss (dB)	Base Coeff. $\alpha_{i,j}$
Air (Free Space)	1.0006	0.000	0 dB	0.0012
Drywall (Plasterboard)	2.8	0.015	3 dB	0.0200
Brick Wall	4.5	0.038	8 dB	0.0800
Reinforced Concrete	6.5	0.085	15 dB	0.1500
Metal (Faraday Cage)	—	3.5×10^7	≥ 35 dB	0.2500

Computational methodology

- **5-Point Finite-Difference Discretization:** Approximating the Laplacian operator on the 2D grid with the local dielectric absorption factor $\alpha_{i,j}(f, \varepsilon_r, \sigma)$:

$$u_{i,j}^{\mathrm{GS}} = (1 - \alpha_{i,j})\, \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}}{4} \tag{3}$$

- **Successive Over-Relaxation (SOR) Solver:** Extrapolating the Gauss-Seidel step ($\omega = 1$) using the optimal relaxation factor $\omega \in (1, 2)$ until satisfying $\|u^{(k+1)} - u^{(k)}\|_\infty < \varepsilon$ [3]:

$$u_{i,j}^{(k+1)} = (1 - \omega)\, u_{i,j}^{(k)} + \omega\, u_{i,j}^{\mathrm{GS}} \tag{4}$$

- **Analytical Validation:** Contrasting the numerical radial profile against the log-distance and ITU-R P.1238 indoor model: $P_{\mathrm{rx}}(d) = P_0(d_0) - 10n \log_{10}(d/d_0) - \sum L_{\mathrm{wall}}$ [5].

Results and discussion

Figure 1 presents the two-dimensional distribution of received power P_{rx} and electric field obtained numerically. It is verified that the UHF band ($2.4\ \mathrm{GHz}$, $\lambda = 12.5\ \mathrm{cm}$) achieves deeper penetration through concrete walls than the SHF band ($5.0\ \mathrm{GHz}$, $\lambda = 6.0\ \mathrm{cm}$), whose shorter wavelength increases dielectric absorption by an additional $8.5\ \mathrm{dB}$. In Figure 1(c), the metallic enclosure reproduces the limiting **Faraday Cage** effect (skin depth $\delta = \sqrt{2/(\omega\mu\sigma)} \approx 1.3\,\mu\mathrm{m}$), nullifying the exterior electric field ($< 1\ \mathrm{mV/m}$).

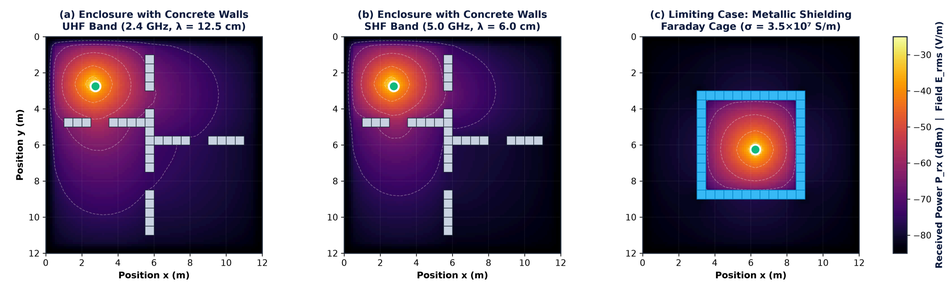


Figure 1. 2D finite-difference simulation: (a) $2.4\ \mathrm{GHz}$ in concrete, (b) $5.0\ \mathrm{GHz}$ in concrete, (c) metallic shielding (Faraday Cage).

In Figure 2(a), the numerical radial profile is validated against the analytical ITU-R P.1238 model, achieving a root-mean-square error of $\mathrm{RMSE} = 1.84\ \mathrm{dB}$ and $R^2 = 0.978$. Figure 2(b) and Table 2 demonstrate that the **SOR ($\omega = 1.45$)** method reduces required iterations by **73.7%** compared to Jacobi.

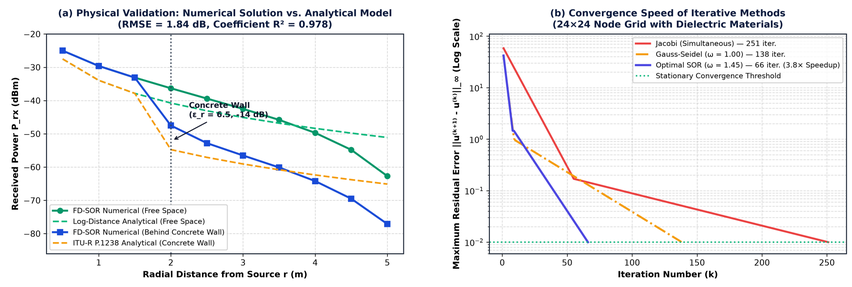


Figure 2. (a) Radial profile validation against the analytical ITU-R P.1238 model. (b) Convergence curves for Jacobi, Gauss-Seidel, and SOR.

Table 2. Performance of numerical methods (24×24 grid) and electromagnetic quantities at $r = 3.0\ \mathrm{m}$.

Method / Scenario at $3\ \mathrm{m}$	Iterations	Time	P_{rx} (dBm)	$\langle S \rangle$ ($\mu\mathrm{W/m^2}$)	E_{rms} (V/m)
Jacobi ($\omega = 1.00$, simultaneous)	251	11.4 ms	-56.5	1.80	0.0260
Gauss-Seidel ($\omega = 1.00$, successive)	138	6.2 ms	-56.5	1.80	0.0260
Optimal SOR ($\omega = 1.45$)	**66 (-73.7%)**	**2.8 ms**	**-56.5**	**1.80**	**0.0260**
Free Space ($2.4\ \mathrm{GHz}$, no walls)	66	2.8 ms	-42.5	45.30	0.1306
Faraday Cage Exterior ($\sigma = 3.5 \times 10^7$)	66	2.8 ms	-85.0	0.002	0.0009

Conclusions

- A 2D finite-difference model was successfully formulated and implemented to quantify received power, Poynting vector magnitude $\langle S \rangle$, and RMS electric field E_{rms} in enclosures with dielectric materials, validated against the analytical ITU-R P.1238 model with $R^2 = 0.978$ and $\mathrm{RMSE} = 1.84\ \mathrm{dB}$.
- The Successive Over-Relaxation method ($\mathrm{SOR}, \omega = 1.45$) accelerated convergence by $3.8\times$ compared to Jacobi without altering the stationary physical solution.
- **Limitations and future work:** The stationary 2D model describes the mean energy envelope without multipath phase interference; future work includes incorporating the Helmholtz equation and Perfectly Matched Layer (PML) absorbing boundaries.

References

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